Practical No. 1

Aim: To implement Basic SQL commands and to access & modify Data using SQL. Create and populate a database using Data Definition Language (DDL) and DML Commands.

Theory:
Structured Query Language (SQL) is a powerful domain-specific language used for managing and manipulating relational databases. It consists of two main categories of commands: Data Definition Language (DDL) for defining the structure of the database, and Data Manipulation Language (DML) for interacting with the data stored in the database. In this document, we will delve into the implementation of basic SQL commands to create and populate a database using DDL and DML commands, while also discussing key concepts and best practices.

DDL commands are used to define, modify, and manage the structure of a database, including tables, indexes, and constraints. Some key DDL commands are:
CREATE, DROP, ALTER, TRUNCATE, SELECT

DML commands are used to interact with the data stored within the database. They include commands for inserting, updating, and deleting data, as well as querying data for retrieval. Some key DML commands are:
UPDATE, INSERT, DELETE

Queries:

1) Change the price of “plate” from 1500 to 2000.

```
mysql> UPDATE PRODUCT
    -> SET PRICE = 2000
    -> WHERE PRODUCT_NO = 121;
Query OK, 1 row affected (0.08 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM PRODUCT;
+------------+----------------+-------+-------------+------------------+-----------------+
| PRODUCT_NO | DESCRIPTION    | PRICE | SUPPLIER_NO | MARKETING_REP_NO | SUPPLY_DEPOT_NO |
+------------+----------------+-------+-------------+------------------+-----------------+
|        120 | REDUCER        |  1200 |        1005 |                5 |               6 |
|        121 | PLATE          |  2000 |        1004 |                3 |               1 |
|        122 | HANDLE         |   700 |        1003 |                2 |               4 |
|        124 | WIDGET REMOVER |   900 |        1005 |                4 |               2 |
|        136 | SIZE WIDGET    |  1000 |        1001 |                1 |               5 |
|        137 | SIZE WIDGET    | 15000 |        1002 |                2 |              16 |
+------------+----------------+-------+-------------+------------------+-----------------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

2) Modify the credit limit to 8000 for those customers who live in “Grange”.

```
mysql> UPDATE CUSTOMER
    -> SET CREDIT_LIMIT = 8000
    -> WHERE ADDRESS = 'GRANGE';
Query OK, 1 row affected (0.12 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM CUSTOMER;
+-------------+--------------+---------------+----------+--------------+
| CUSTOMER_NO | NAME         | ADDRESS       | DEPOT_NO | CREDIT_LIMIT |
+-------------+--------------+---------------+----------+--------------+
|          10 | GARRY SMITH  | BRIXTON       |        6 |         1000 |
|          20 | PATEL        | GRANGE        |        1 |         8000 |
|          30 | DRAKE        | BRIXTON       |        4 |         7000 |
|          40 | BOB SMITH    | LONDON        |        2 |        10000 |
|          50 | JAMES        | GRANCE        |        3 |         5000 |
|          60 | NORTON       | SAN FRANSISCO |        5 |        17000 |
|          70 | JOHN MICHAEL | EUROPE        |       16 |         8000 |
+-------------+--------------+---------------+----------+--------------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

3) Change the size of the customer address to 30.

```
mysql> ALTER TABLE CUSTOMER
    -> MODIFY ADDRESS varchar(30);
Query OK, 7 rows affected (0.97 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CUSTOMER;
+-------------+--------------+---------------+----------+--------------+
| CUSTOMER_NO | NAME         | ADDRESS       | DEPOT_NO | CREDIT_LIMIT |
+-------------+--------------+---------------+----------+--------------+
|          10 | GARRY SMITH  | BRIXTON       |        6 |         1000 |
|          20 | PATEL        | GRANGE        |        1 |         8000 |
|          30 | DRAKE        | BRIXTON       |        4 |         7000 |
|          40 | BOB SMITH    | LONDON        |        2 |        10000 |
|          50 | JAMES        | GRANCE        |        3 |         5000 |
|          60 | NORTON       | SAN FRANSISCO |        5 |        17000 |
|          70 | JOHN MICHAEL | EUROPE        |       16 |         8000 |
+-------------+--------------+---------------+----------+--------------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

4) Create a table cust1 with the attributes and formats
Customer_no number (10)
Name varchar2 (20)
Address varchar2 (20)
Rep_no number (10)

```
mysql> CREATE TABLE CUST1(
    -> CUSTOMER_NO int(10) PRIMARY KEY NOT NULL,
    -> NAME varchar(20) NOT NULL,
    -> ADDRESS varchar(20) NOT NULL,
    -> REP_NO int(10) NOT NULL
    -> );
Query OK, 0 rows affected (0.39 sec)

mysql> #202203103510097
```

5) Add a new field email id in the cust1 table.

```
mysql> ALTER TABLE CUST1
    -> ADD EMAIL_ID varchar(20);
Query OK, 0 rows affected (0.62 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> #202203103510097
```

6) Display the structure of the cust1 table.

```
mysql> DESCRIBE CUST1;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| CUSTOMER_NO | int(10)     | NO   | PRI | NULL    |       |
| NAME        | varchar(20) | NO   |     | NULL    |       |
| ADDRESS     | varchar(20) | NO   |     | NULL    |       |
| REP_NO      | int(10)     | NO   |     | NULL    |       |
| EMAIL_ID    | varchar(20) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> #202203103510097
```

7) Display the content of the cust1 table.

```
mysql> SELECT * FROM CUST1;
+-------------+--------------+---------------+--------+----------+
| CUSTOMER_NO | NAME         | ADDRESS       | REP_NO | EMAIL_ID |
+-------------+--------------+---------------+--------+----------+
|          10 | GARRY SMITH  | BRIXTON       |      6 | NULL     |
|          20 | PATEL        | GRANGE        |      1 | NULL     |
|          30 | DRAKE        | BRIXTON       |      4 | NULL     |
|          40 | BOB SMITH    | LONDON        |      2 | NULL     |
|          50 | JAMES        | GRANCE        |      3 | NULL     |
|          60 | NORTON       | SAN FRANSISCO |      5 | NULL     |
|          70 | JOHN MICHAEL | EUROPE        |      2 | NULL     |
+-------------+--------------+---------------+--------+----------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

8) Delete details of customer no 2 from cust1 table.

```
mysql> DELETE FROM CUST1
    -> WHERE CUSTOMER_NO = 20;
Query OK, 1 row affected (0.07 sec)

mysql> SELECT * FROM CUST1;
+-------------+--------------+---------------+--------+----------+
| CUSTOMER_NO | NAME         | ADDRESS       | REP_NO | EMAIL_ID |
+-------------+--------------+---------------+--------+----------+
|          10 | GARRY SMITH  | BRIXTON       |      6 | NULL     |
|          30 | DRAKE        | BRIXTON       |      4 | NULL     |
|          40 | BOB SMITH    | LONDON        |      2 | NULL     |
|          50 | JAMES        | GRANCE        |      3 | NULL     |
|          60 | NORTON       | SAN FRANSISCO |      5 | NULL     |
|          70 | JOHN MICHAEL | EUROPE        |      2 | NULL     |
+-------------+--------------+---------------+--------+----------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

9) Delete email id field from cust1 table.

```
mysql> ALTER TABLE CUST1
    -> DROP COLUMN EMAIL_ID;
Query OK, 0 rows affected (0.96 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CUST1;
+-------------+--------------+---------------+--------+
| CUSTOMER_NO | NAME         | ADDRESS       | REP_NO |
+-------------+--------------+---------------+--------+
|          10 | GARRY SMITH  | BRIXTON       |      6 |
|          30 | DRAKE        | BRIXTON       |      4 |
|          40 | BOB SMITH    | LONDON        |      2 |
|          50 | JAMES        | GRANCE        |      3 |
|          60 | NORTON       | SAN FRANSISCO |      5 |
|          70 | JOHN MICHAEL | EUROPE        |      2 |
+-------------+--------------+---------------+--------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

10) Delete all the data rows from cust1 and look at the contents again.

```
mysql> DELETE FROM CUST1;
Query OK, 6 rows affected (0.05 sec)

mysql> SELECT * FROM CUST1;
Empty set (0.00 sec)

mysql> #202203103510097
```

11) Delete the table cust1 and then try to look at its contents again.

```
mysql> DROP TABLE CUST1;
Query OK, 0 rows affected (0.18 sec)

mysql> SELECT * FROM CUST1;
ERROR 1146 (42S02): Table 'GIDEON.CUST1' doesn't exist
mysql> #202203103510097
```

12) List the customer numbers (customer_no) and names (name) of all customers

```
mysql> SELECT CUSTOMER_NO, NAME FROM CUSTOMER;
+-------------+--------------+
| CUSTOMER_NO | NAME         |
+-------------+--------------+
|          10 | GARRY SMITH  |
|          20 | PATEL        |
|          30 | DRAKE        |
|          40 | BOB SMITH    |
|          50 | JAMES        |
|          60 | NORTON       |
|          70 | JOHN MICHAEL |
+-------------+--------------+
7 rows in set (0.01 sec)

mysql> #202203103510097
```

13) List all details of the product with a product number (product_no) of 121 and 136. (use Or).

```
mysql> SELECT * FROM PRODUCT WHERE PRODUCT_NO = 121 OR PRODUCT_NO = 136;
+------------+-------------+-------+-------------+-----------------+----------------+
| PRODUCT_NO | DESCRIPTION | PRICE | SUPPLIER_NO | MARKETING_REP_NO | SUPPLY_DEPOT_NO |
+------------+-------------+-------+-------------+-----------------+----------------+
|        121 | PLATE       |  2000 |        1004 |               3 |              1 |
|        136 | SIZE WIDGET |  1000 |        1001 |               1 |              5 |
+------------+-------------+-------+-------------+-----------------+----------------+
2 rows in set (0.01 sec)

mysql> #202203103510097
```

14) List all details of depots with rep 5 as their rep(rep_no).

```
mysql> SELECT * FROM DEPOT WHERE REP_NO = 5;
+----------+----------+---------+--------+
| DEPOT_NO | LOCATION | ADDRESS | REP_NO |
+----------+----------+---------+--------+
|        5 | WALES    | UK      |      5 |
+----------+----------+---------+--------+
1 row in set (0.03 sec)

mysql> #202203103510097
```

15) List the product number (product_no) and description only of all products from supplier number 1005 (supplier_no).

```
mysql> SELECT PRODUCT_NO, DESCRIPTION FROM PRODUCT WHERE SUPPLIER_NO = 1005;
+------------+----------------+
| PRODUCT_NO | DESCRIPTION    |
+------------+----------------+
|        120 | REDUCER        |
|        124 | WIDGET REMOVER |
+------------+----------------+
2 rows in set (0.00 sec)

mysql> #202203103510097
```

16) List the sales rep number (rep_no), depot number and address for depots located at NORTH and address is UK.

```
mysql> SELECT REP_NO, DEPOT_NO, ADDRESS FROM DEPOT WHERE LOCATION = 'NORTH' and ADDRESS = 'UK';
+--------+----------+---------+
| REP_NO | DEPOT_NO | ADDRESS |
+--------+----------+---------+
|      1 |        1 | UK      |
+--------+----------+---------+
1 row in set (0.00 sec)

mysql> #202203103510097
```

Conclusion: SQL is a fundamental tool for managing databases, enabling the creation, modification, and retrieval of data. DDL commands are used to define the structure of the database, while DML commands handle data manipulation. By following best practices and understanding the core concepts of SQL, developers can create efficient, organized, and secure databases that cater to their data management needs.

```
mysql> CREATE TABLE SUPPLIER(
    -> SUPPLIER_NO int PRIMARY KEY NOT NULL,
    -> NAME varchar(50) NOT NULL,
    -> ADDRESS varchar(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.37 sec)

mysql> INSERT INTO SUPPLIER VALUE(1001,'MICHAEL','BASILDON'),
    -> (1002,'RINGWORLD','GERMANY'),
    -> (1003,'BABYLON','LONDON'),
    -> (1004,'JOHN','BASILDON'),
    -> (1005,'SMITH','GERMANY');
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM SUPPLIER;
+-------------+-----------+----------+
| SUPPLIER_NO | NAME      | ADDRESS  |
+-------------+-----------+----------+
|        1001 | MICHAEL   | BASILDON |
|        1002 | RINGWORLD | GERMANY  |
|        1003 | BABYLON   | LONDON   |
|        1004 | JOHN      | BASILDON |
|        1005 | SMITH     | GERMANY  |
+-------------+-----------+----------+
5 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE PRODUCT(
    -> PRODUCT_NO int PRIMARY KEY NOT NULL,
    -> DESCRIPTION varchar(50) NOT NULL,
    -> PRICE int NOT NULL,
    -> SUPPLIER_NO int NOT NULL,
    -> MARKETING_REP_NO int NOT NULL,
    -> SUPPLY_DEPOT_NO int NOT NULL
    -> );
Query OK, 0 rows affected (0.36 sec)

mysql> INSERT INTO PRODUCT VALUE(120,'REDUCER',1200,1005,5,6),
    -> (121,'PLATE',1500,1004,3,1),
    -> (122,'HANDLE',700,1003,2,4),
    -> (124,'WIDGET REMOVER',900,1005,4,2),
    -> (136,'SIZE WIDGET',1000,1001,1,5),
    -> (137,'SIZE WIDGET',15000,1002,2,16);
Query OK, 6 rows affected (0.05 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM PRODUCT;
+------------+----------------+-------+-------------+------------------+-----------------+
| PRODUCT_NO | DESCRIPTION    | PRICE | SUPPLIER_NO | MARKETING_REP_NO | SUPPLY_DEPOT_NO |
+------------+----------------+-------+-------------+------------------+-----------------+
|        120 | REDUCER        |  1200 |        1005 |                5 |               6 |
|        121 | PLATE          |  1500 |        1004 |                3 |               1 |
|        122 | HANDLE         |   700 |        1003 |                2 |               4 |
|        124 | WIDGET REMOVER |   900 |        1005 |                4 |               2 |
|        136 | SIZE WIDGET    |  1000 |        1001 |                1 |               5 |
|        137 | SIZE WIDGET    | 15000 |        1002 |                2 |              16 |
+------------+----------------+-------+-------------+------------------+-----------------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE SALESREP(
    -> REP_NO int PRIMARY KEY NOT NULL,
    -> NAME varchar(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.33 sec)

mysql> INSERT INTO SALESREP VALUE(1,'MIKE'),
    -> (2,'FRED'),
    -> (3,'ALI'),
    -> (4,'SAM'),
    -> (5,'BILL ADAMS'),
    -> (6,'SAM');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM SALESREP;
+--------+------------+
| REP_NO | NAME       |
+--------+------------+
|      1 | MIKE       |
|      2 | FRED       |
|      3 | ALI        |
|      4 | SAM        |
|      5 | BILL ADAMS |
|      6 | SAM        |
+--------+------------+
6 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE DEPOT(
    -> DEPOT_NO int PRIMARY KEY NOT NULL,
    -> LOCATION varchar(50) NOT NULL,
    -> ADDRESS varchar(50) NOT NULL,
    -> REP_NO int NOT NULL
    -> );
Query OK, 0 rows affected (0.30 sec)

mysql> INSERT INTO DEPOT VALUE(1,'NORTH','UK',1),
    -> (2,'SOUTH','USA',2),
    -> (3,'LONDON WEST','USA',3),
    -> (4,'EAST','NZ',4),
    -> (5,'WALES','UK',5),
    -> (6,'NORTH','KENYA',6),
    -> (16,'SOUTH','UK',2);
Query OK, 7 rows affected (0.05 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM DEPOT;
+----------+-------------+---------+--------+
| DEPOT_NO | LOCATION    | ADDRESS | REP_NO |
+----------+-------------+---------+--------+
|        1 | NORTH       | UK      |      1 |
|        2 | SOUTH       | USA     |      2 |
|        3 | LONDON WEST | USA     |      3 |
|        4 | EAST        | NZ      |      4 |
|        5 | WALES       | UK      |      5 |
|        6 | NORTH       | KENYA   |      6 |
|       16 | SOUTH       | UK      |      2 |
+----------+-------------+---------+--------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE CUSTOMER(
    -> CUSTOMER_NO int PRIMARY KEY NOT NULL,
    -> NAME varchar(50) NOT NULL,
    -> ADDRESS varchar(50) NOT NULL,
    -> DEPOT_NO int NOT NULL,
    -> CREDIT_LIMIT int NOT NULL
    -> );
Query OK, 0 rows affected (0.51 sec)

mysql> INSERT INTO CUSTOMER VALUE(10,'GARRY SMITH','BRIXTON',6,1000),
    -> (20,'PATEL','GRANGE',1,4000),
    -> (30,'DRAKE','BRIXTON',4,7000),
    -> (40,'BOB SMITH','LONDON',2,10000),
    -> (50,'JAMES','GRANCE',3,5000),
    -> (60,'NORTON','SAN FRANSISCO',5,17000),
    -> (70,'JOHN MICHAEL','EUROPE',16,8000);
Query OK, 7 rows affected (0.04 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CUSTOMER;
+-------------+--------------+---------------+----------+--------------+
| CUSTOMER_NO | NAME         | ADDRESS       | DEPOT_NO | CREDIT_LIMIT |
+-------------+--------------+---------------+----------+--------------+
|          10 | GARRY SMITH  | BRIXTON       |        6 |         1000 |
|          20 | PATEL        | GRANGE        |        1 |         4000 |
|          30 | DRAKE        | BRIXTON       |        4 |         7000 |
|          40 | BOB SMITH    | LONDON        |        2 |        10000 |
|          50 | JAMES        | GRANCE        |        3 |         5000 |
|          60 | NORTON       | SAN FRANSISCO |        5 |        17000 |
|          70 | JOHN MICHAEL | EUROPE        |       16 |         8000 |
+-------------+--------------+---------------+----------+--------------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE CORDER(
    -> CORDER_NO int PRIMARY KEY NOT NULL,
    -> CUSTOMER_NO int NOT NULL,
    -> DATE_PLACED varchar(50) NOT NULL,
    -> DATE_DELIVERED varchar(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.35 sec)

mysql> INSERT INTO CORDER VALUE(200,20,'01-JAN-1993','04-JAN-1993'),
    -> (201,40,'17-JAN-1993','20-JAN-1993'),
    -> (202,20,'01-JAN-1993','04-JAN-1993'),
    -> (203,30,'02-FEB-1995','05-JAN-1995'),
    -> (204,10,'13-MAR-1996','16-MAR-1996'),
    -> (205,70,'31-JAN-1993','03-FEB-1993'),
    -> (206,40,'01-JAN-1994','04-JAN-1994'),
    -> (207,20,'02-AUG-1994','05-AUG-1994');
Query OK, 8 rows affected (0.07 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CORDER;
+-----------+-------------+-------------+----------------+
| CORDER_NO | CUSTOMER_NO | DATE_PLACED | DATE_DELIVERED |
+-----------+-------------+-------------+----------------+
|       200 |          20 | 01-JAN-1993 | 04-JAN-1993    |
|       201 |          40 | 17-JAN-1993 | 20-JAN-1993    |
|       202 |          20 | 01-JAN-1993 | 04-JAN-1993    |
|       203 |          30 | 02-FEB-1995 | 05-JAN-1995    |
|       204 |          10 | 13-MAR-1996 | 16-MAR-1996    |
|       205 |          70 | 31-JAN-1993 | 03-FEB-1993    |
|       206 |          40 | 01-JAN-1994 | 04-JAN-1994    |
|       207 |          20 | 02-AUG-1994 | 05-AUG-1994    |
+-----------+-------------+-------------+----------------+
8 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE ONLINE(
    -> CORDER_NO int PRIMARY KEY NOT NULL,
    -> PRODUCT_NO int NOT NULL,
    -> QUANTITY int NOT NULL
    -> );
Query OK, 0 rows affected (0.37 sec)

mysql> INSERT INTO ONLINE VALUE(200,120,5),
    -> (201,121,10),
    -> (202,120,5),
    -> (203,122,20),
    -> (204,136,30),
    -> (205,124,15),
    -> (206,136,30);
Query OK, 7 rows affected (0.05 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM ONLINE;
+-----------+------------+----------+
| CORDER_NO | PRODUCT_NO | QUANTITY |
+-----------+------------+----------+
|       200 |        120 |        5 |
|       201 |        121 |       10 |
|       202 |        120 |        5 |
|       203 |        122 |       20 |
|       204 |        136 |       30 |
|       205 |        124 |       15 |
|       206 |        136 |       30 |
+-----------+------------+----------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```

```
mysql> CREATE TABLE STOCK(
    -> DEPOT_NO int PRIMARY KEY NOT NULL,
    -> PRODUCT_NO int NOT NULL,
    -> QUANTITY int NOT NULL,
    -> RACK int NOT NULL,
    -> BIN_NO int NOT NULL
    -> );
Query OK, 0 rows affected (0.39 sec)

mysql> INSERT INTO STOCK VALUE(1,120,50,1,1),
    -> (2,137,100,10,2),
    -> (3,136,40,2,3),
    -> (4,120,60,7,1),
    -> (5,121,90,5,4),
    -> (6,124,120,4,7),
    -> (16,122,80,10,8);
Query OK, 7 rows affected (0.05 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM STOCK;
+----------+------------+----------+------+--------+
| DEPOT_NO | PRODUCT_NO | QUANTITY | RACK | BIN_NO |
+----------+------------+----------+------+--------+
|        1 |        120 |       50 |    1 |      1 |
|        2 |        137 |      100 |   10 |      2 |
|        3 |        136 |       40 |    2 |      3 |
|        4 |        120 |       60 |    7 |      1 |
|        5 |        121 |       90 |    5 |      4 |
|        6 |        124 |      120 |    4 |      7 |
|       16 |        122 |       80 |   10 |      8 |
+----------+------------+----------+------+--------+
7 rows in set (0.00 sec)

mysql> #202203103510097
```